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Luiz

SEL0337 Projetos em Sistemas Embarcados

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Historico pratica 3 pt1:

109 ls
110 echo 18 > /sys/class/gpio/export
111 echo out > /sys/class/gpio/gpio18/direction
112 sudo rpi-update cac01bed1224743104cb2a4103605f269f207b1a
113 y
114 pinout
115 sudo rpi-update cac01bed1224743104cb2a4103605f269f207b1a
116 y
117 N
118 sudo apt-get install python
119 sudo
120 apt-get install python3-pip
121 sudo
122 apt-get install python3-pip
123 sudo apt-get install python3-pip
124 python3 -m pip install - -user virtualenv
125 sudo apt install python3-venv -y
126 python3 -m venv 4265
127 source 4265/bin/activate
128 deactivate
129 source 4265/bin/activate
130 python
131 cd 4265
132 pip freeze
133 pip3 install <>
134 pip3
135 install <>
136 install <gpiozero, RPi.GPIO, lgpio>
137 install gpiozero
138 install --help
139 quit() python
140 pip freeze
141 python --version
142 pip --version
143 cd 4265
144 ls
145 cd

```
146  ls
147  cd 4265
148  ls
149  cd
150  source 4265/bin/activate
151  ls
152  deactivate
153  source 4265/bin/activate
154  deactivate
155  pip freeze
156  source 4265/bin/activate
157  pip freeze
158  pip3 install <piozero>
159  pip3 install gpiozero
160  pip freeze
161  pip3 install RPi.GPIO
162  pip freeze
163  pip3 install lgpio
164  pip freeze
165  pinout
166  cd 4265
167  nano blink_gpiozero.py
168  python3 blink_gpiozero.py
169  nano blink_gpiozero.py
170  python3 blink_gpiozero.py
171  history > pratica3parte1.txt
```

Blink_gpiozero.py:

```python
from gpiozero import LED
from time import sleep

led = LED(18)

while True:
	led.on()
	print("LED on")
	sleep(1)
	led.off()
	sleep(1)
	print("LED off")
```

cronometro_casa.py:

```python
import time
```

```
import os

segundos = int(input("Digite o número de segundos: "))

for i in range(segundos,0,-1):
#calcula minutos e segundos
        minutos=i//60
        resto_segundos=i%60
#mostra cronometro
        print(f"Cronômetro: {minutos:02}:{resto_segundos:02}")
#espera 1 segundo
        time.sleep(1)


print("Tempo finalizado.")
```

Botao.py: (código de botão que aciona LED)

```
import RPi.GPIO as GPIO
from gpiozero  import LED

GPIO.setmode(GPIO.BCM)

red=LED(24)

def button(pin):
      if GPIO.input(pin):
            print("released")
            red.off()
      else:
            print("pressed")
            red.on()
GPIO.setup(18, GPIO.IN, GPIO.PUD_UP)
GPIO.add_event_detect(18, GPIO.BOTH, callback =button, bouncetime=200)
try:
      while True:
                  pass
except KeyboardInterrupt:
      GPIO.cleanup()
```

PWM.py: código que aciona LED variando em PWM:

```
from gpiozero import PWMLED
from time import sleep

led = PWMLED(24)
```

```
while True:
      led.value = 0
      sleep(0.5)
      led.value = 0.25
      sleep(0.5)
      led.value = 0.5
      sleep(0.5)
      led.value = 0.75
      sleep(0.5)
      led.value = 1
      sleep(0.5)
```

sensormovimento.py: (código que implementa sensor de movimento)

```
from gpiozero import MotionSensor , LED
from signal import pause

pir = MotionSensor(4)
led = LED(24)

pir.when_motion = led.on
pir.when_no_motion = led.off

pause()
```

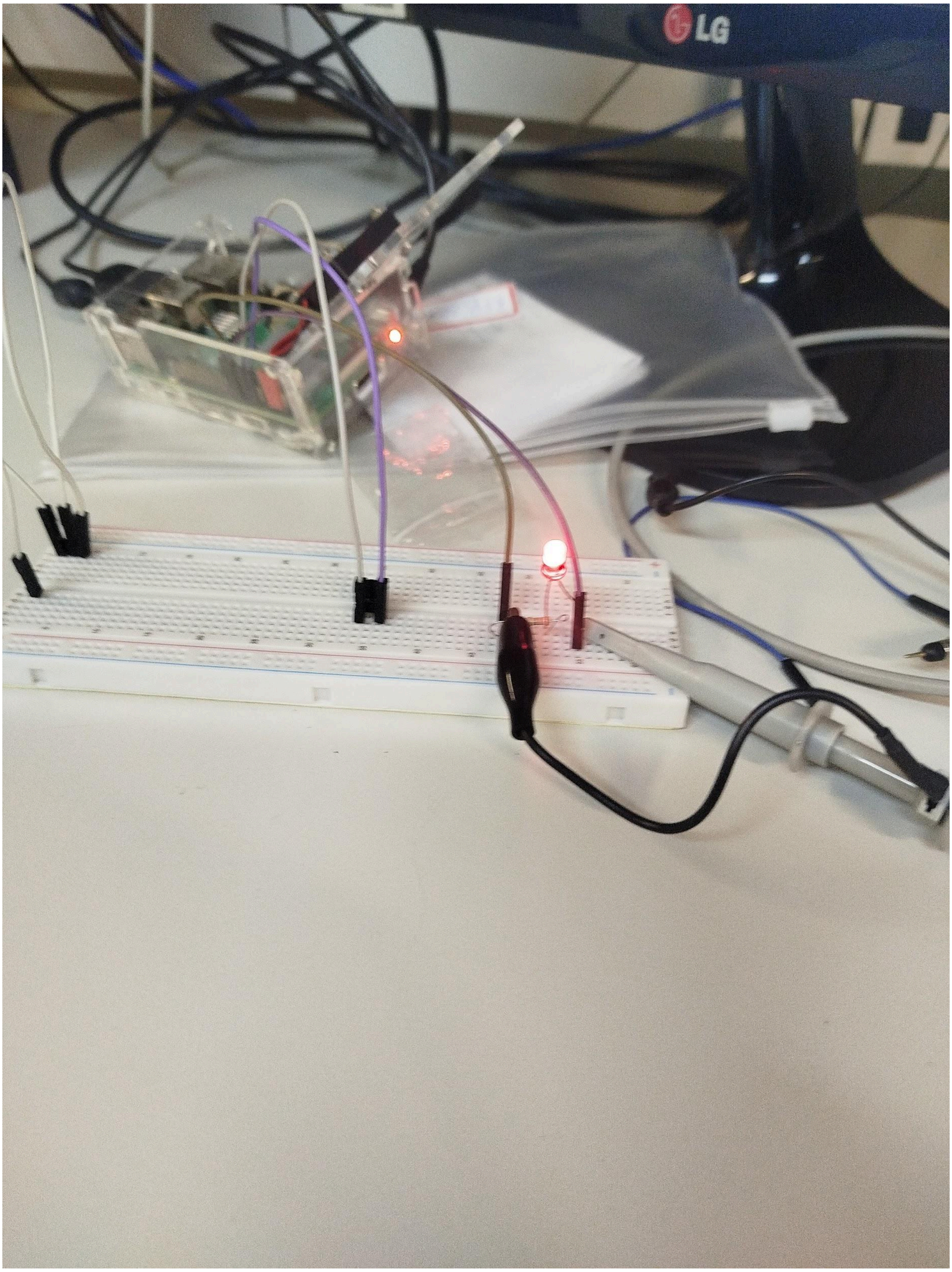
pythreads.py: (código que implementa threads com gpio)

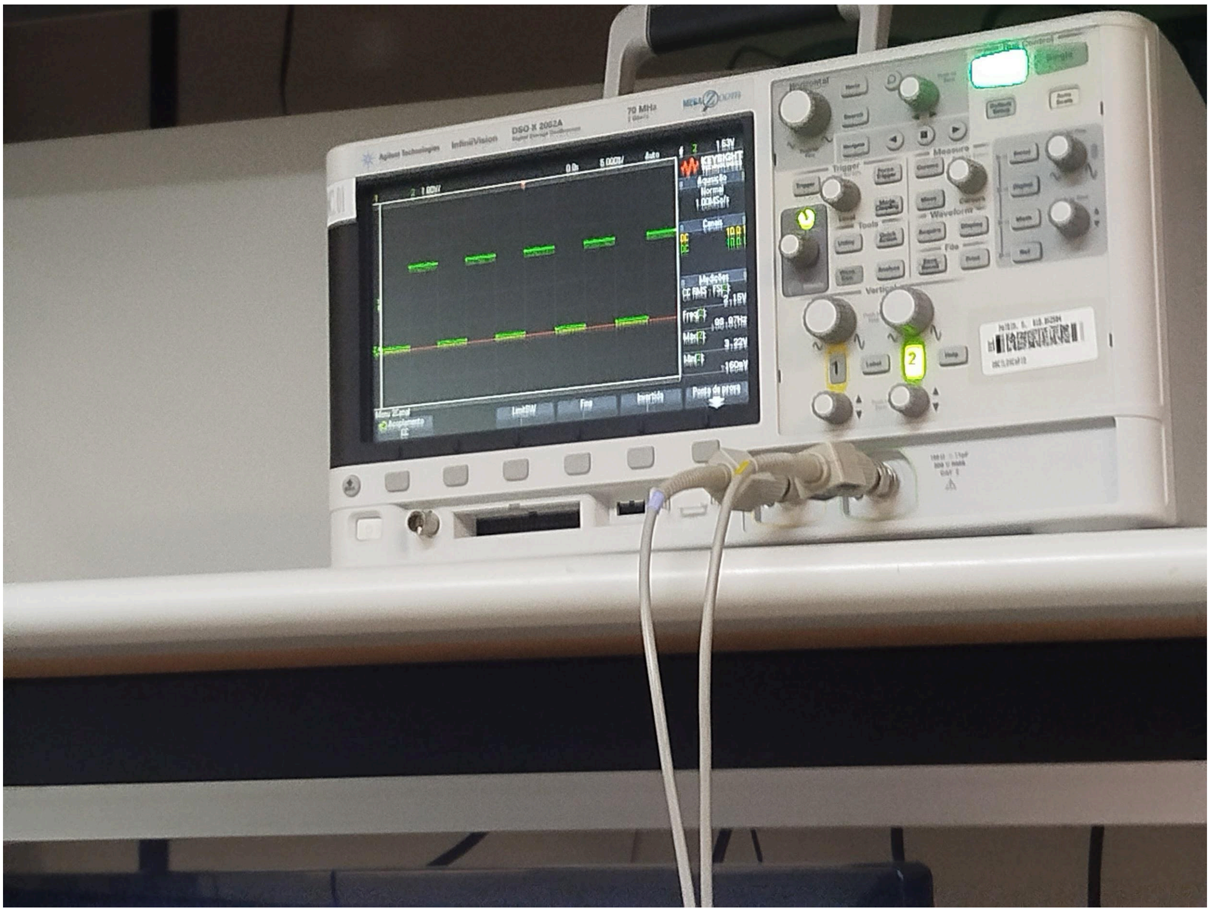
```
import RPi.GPIO as GPIO
import time
import random
from multiprocessing import Process
import os

def gpio_blink():
    print(f"Processo de Blink LED iniciado com PID: {os.getpid()}")
    GPIO.setmode(GPIO.BOARD)
    GPIO.setup(18, GPIO.OUT)
    while True:
        GPIO.output(18, GPIO.HIGH)
        print("LED aceso")
        time.sleep(1)
        GPIO.output(18, GPIO.LOW)
        print("LED apagado")
        time.sleep(1)

def random_count():
    print(f"Processo de Contagem Aleatória iniciado com PID:
{os.getpid()}")
    while True:
        count = random.randint(1, 100)
```

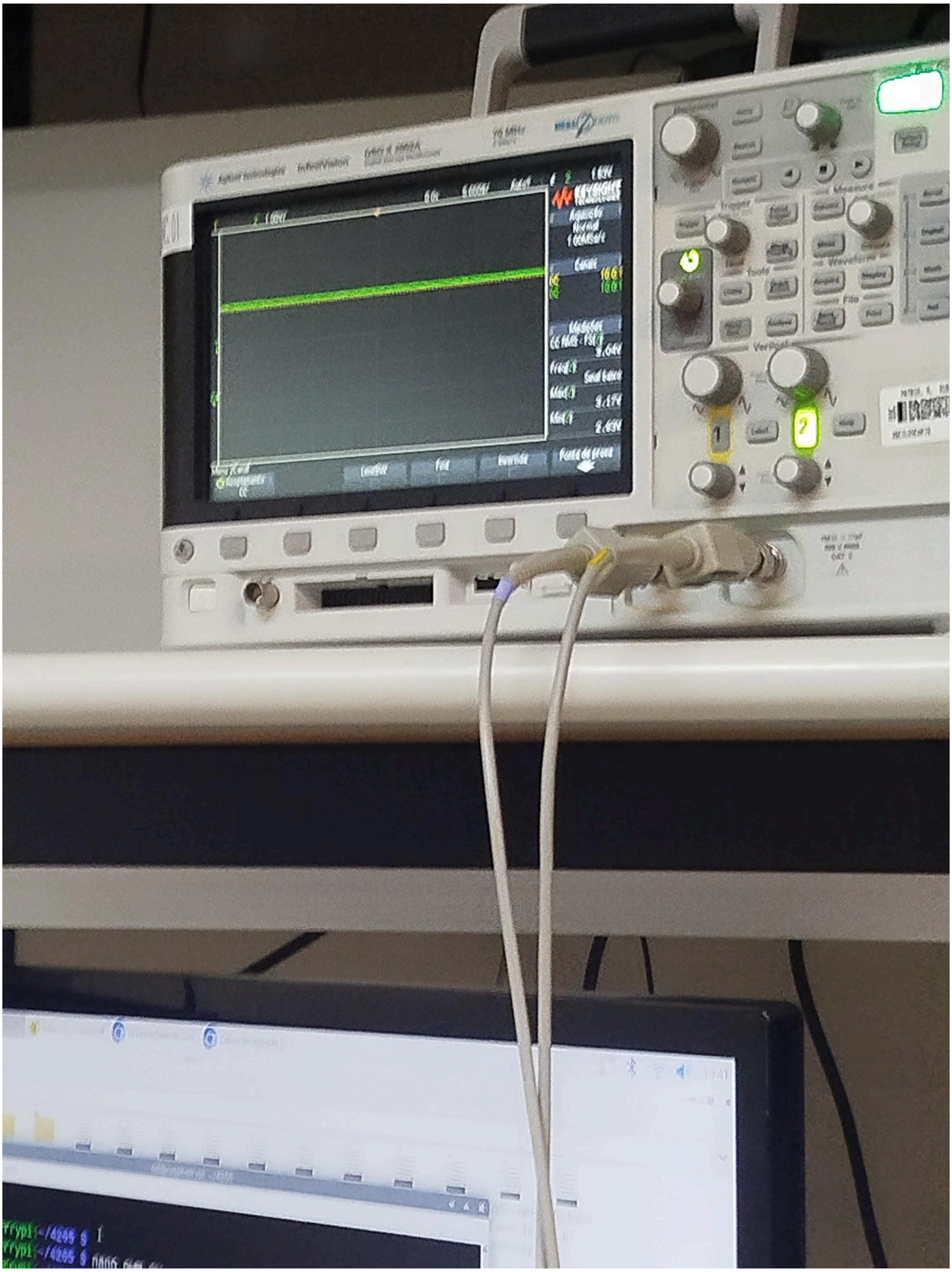
```
        print(f"Contagem aleatória: {count}")
        time.sleep(2)

if __name__ == '__main__':
    try:
        process1 = Process(target=gpio_blink)
        process2 = Process(target=random_count)

        process1.start()
        process2.start()

        process1.join()
        process2.join()
    except KeyboardInterrupt:
        print("Processos interrompidos.")
    finally:
        GPIO.cleanup()
```

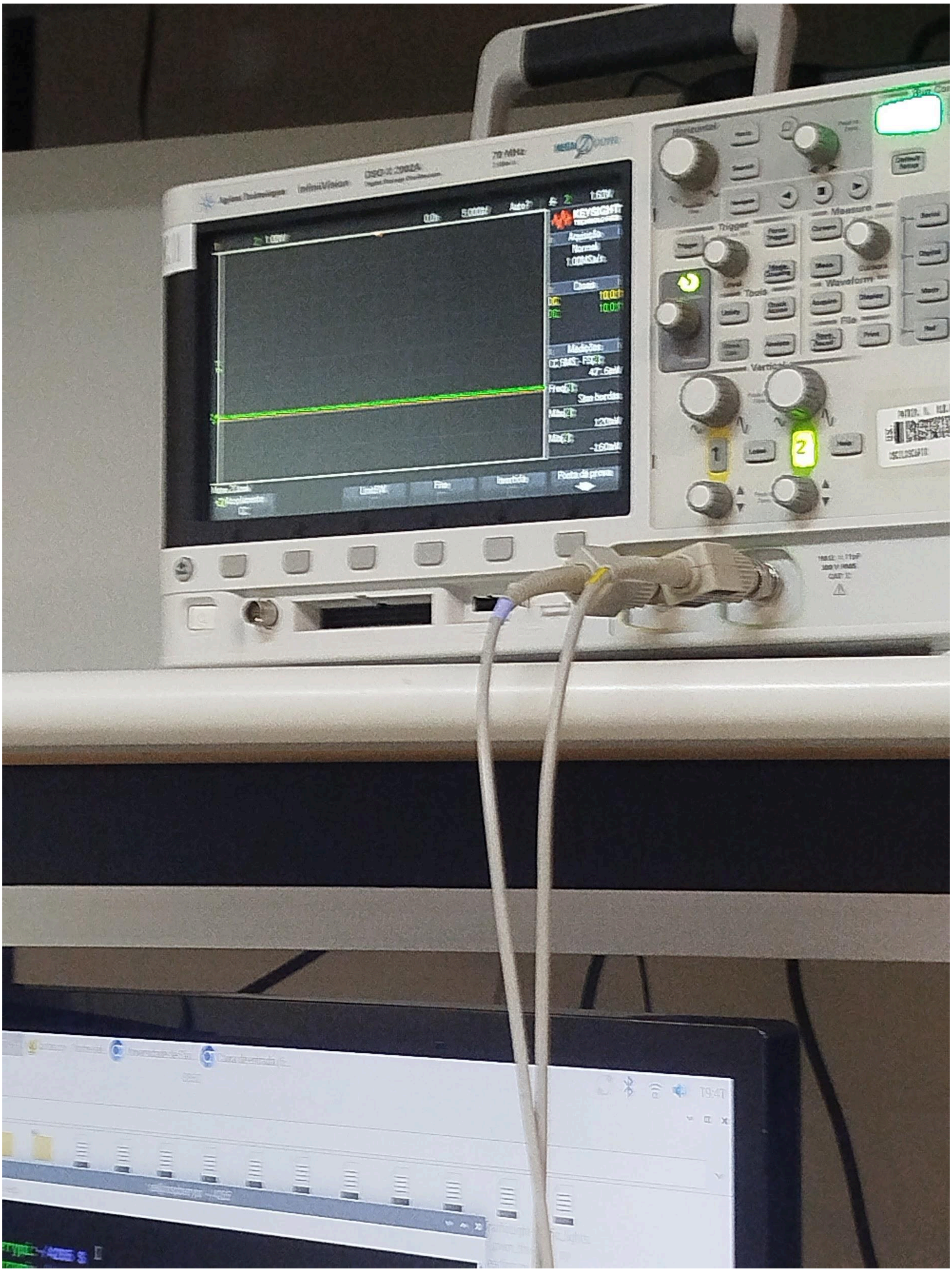
DOCUMENTAÇÃO EM FOTOS DOS EXEMPLOS DA PRÁTICA:

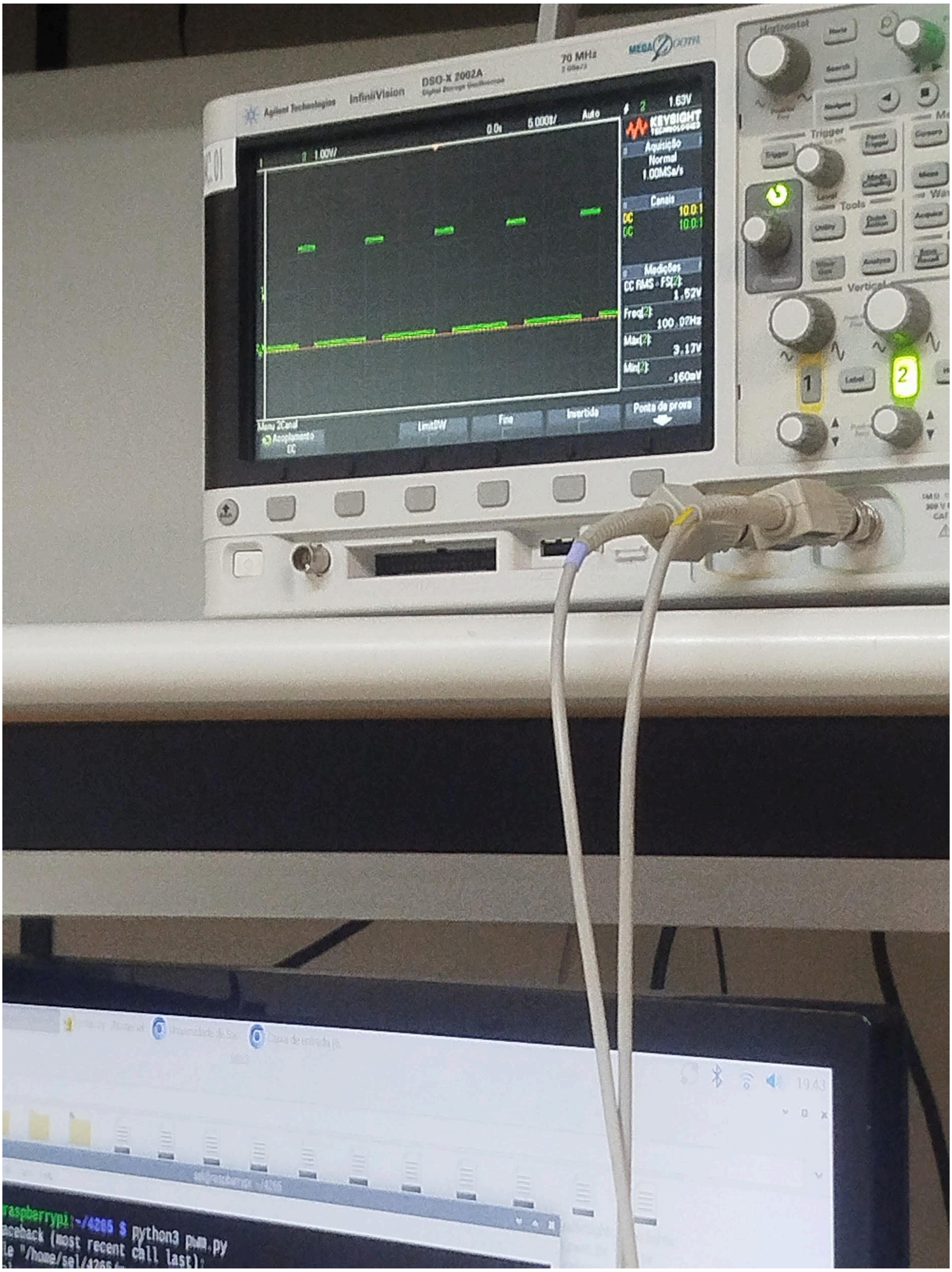
IMAGENS: IMPLEMENTAÇÃO DO PWM.py:

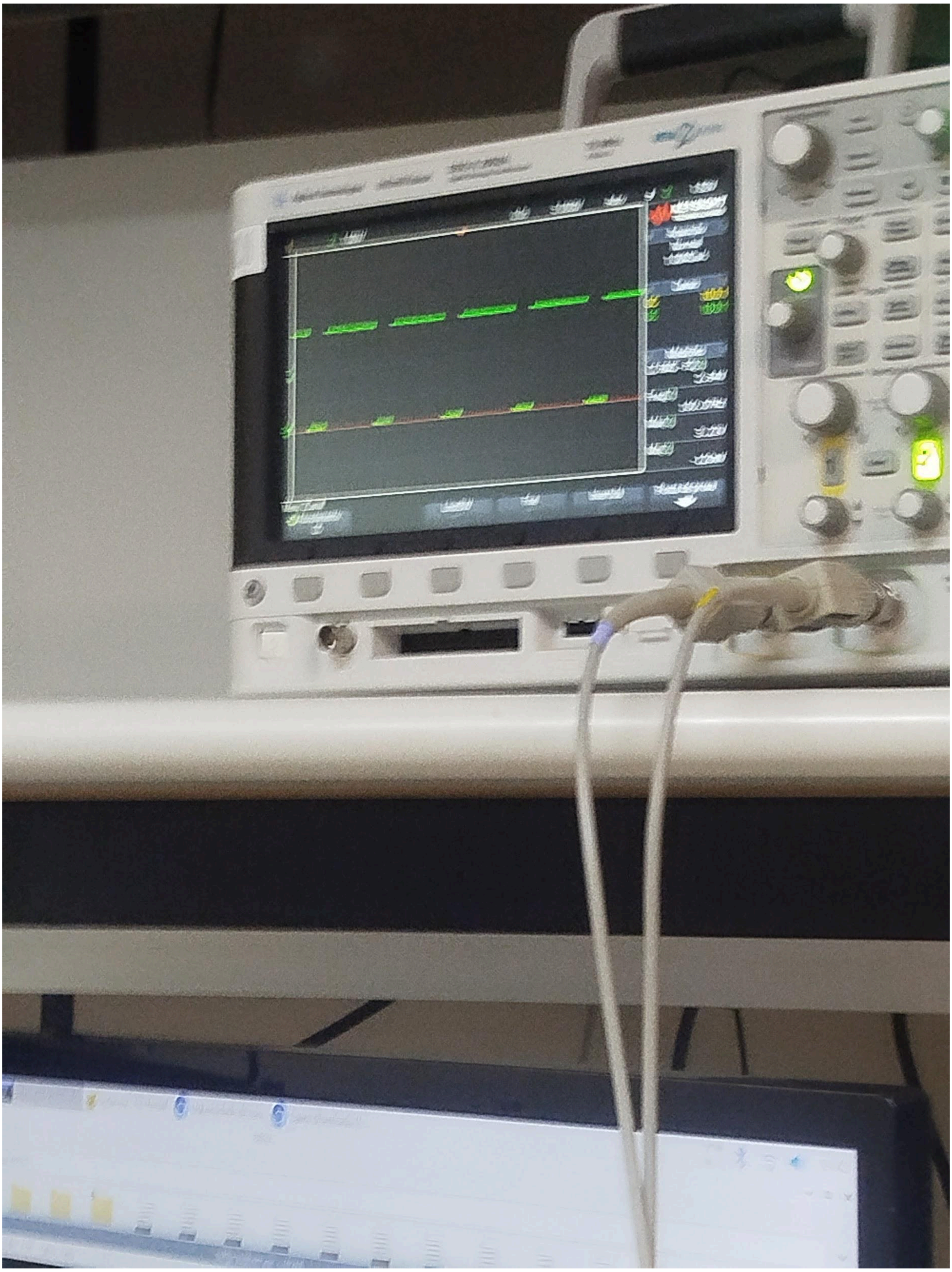

LG


DSO X 2002A
70 MHz
KEYSIGHT

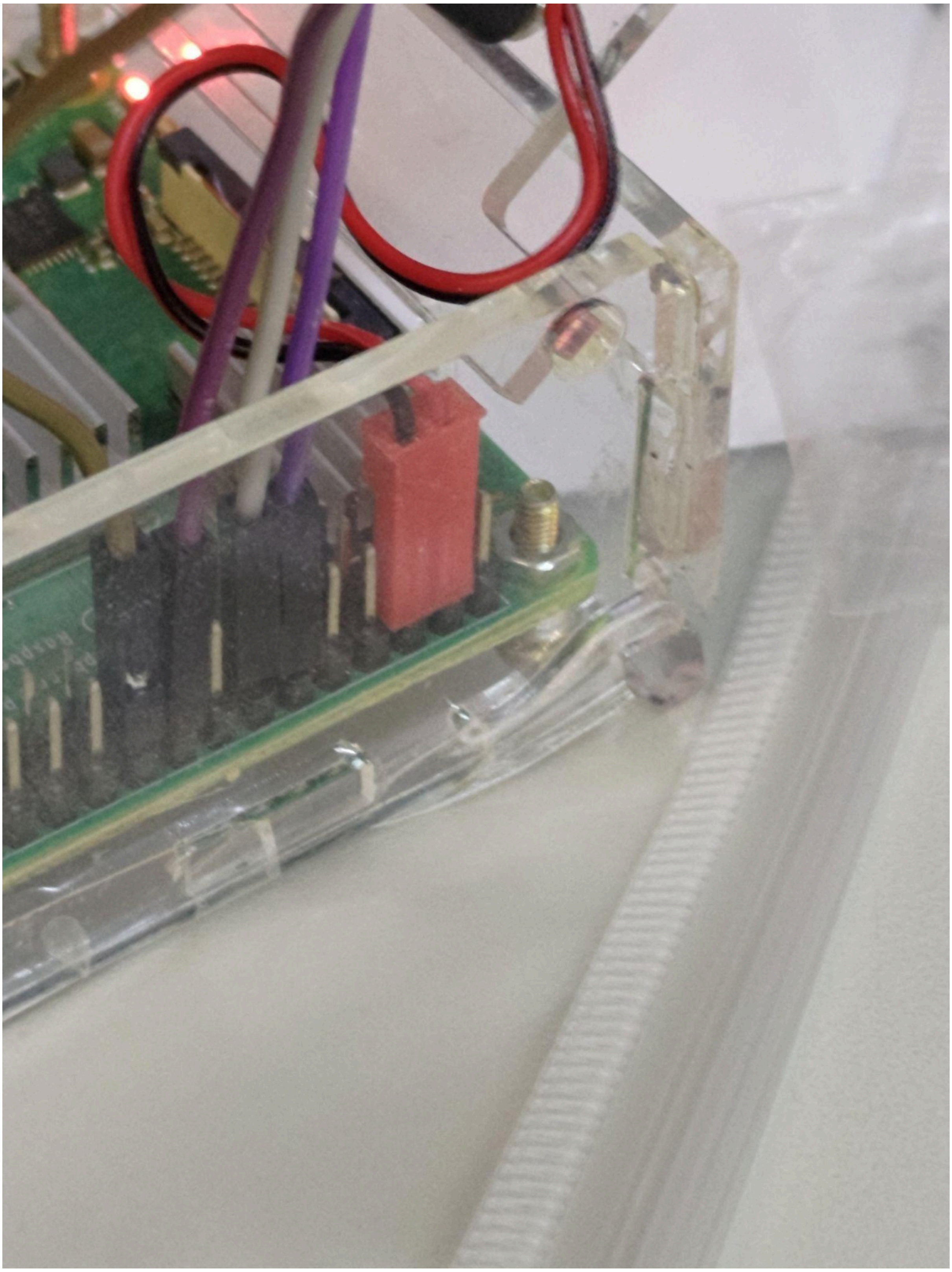

InfiniiVision
KEYSIGHT
1.63V
9.64V
9.17V
2.63V
Fina
Ponta de prova

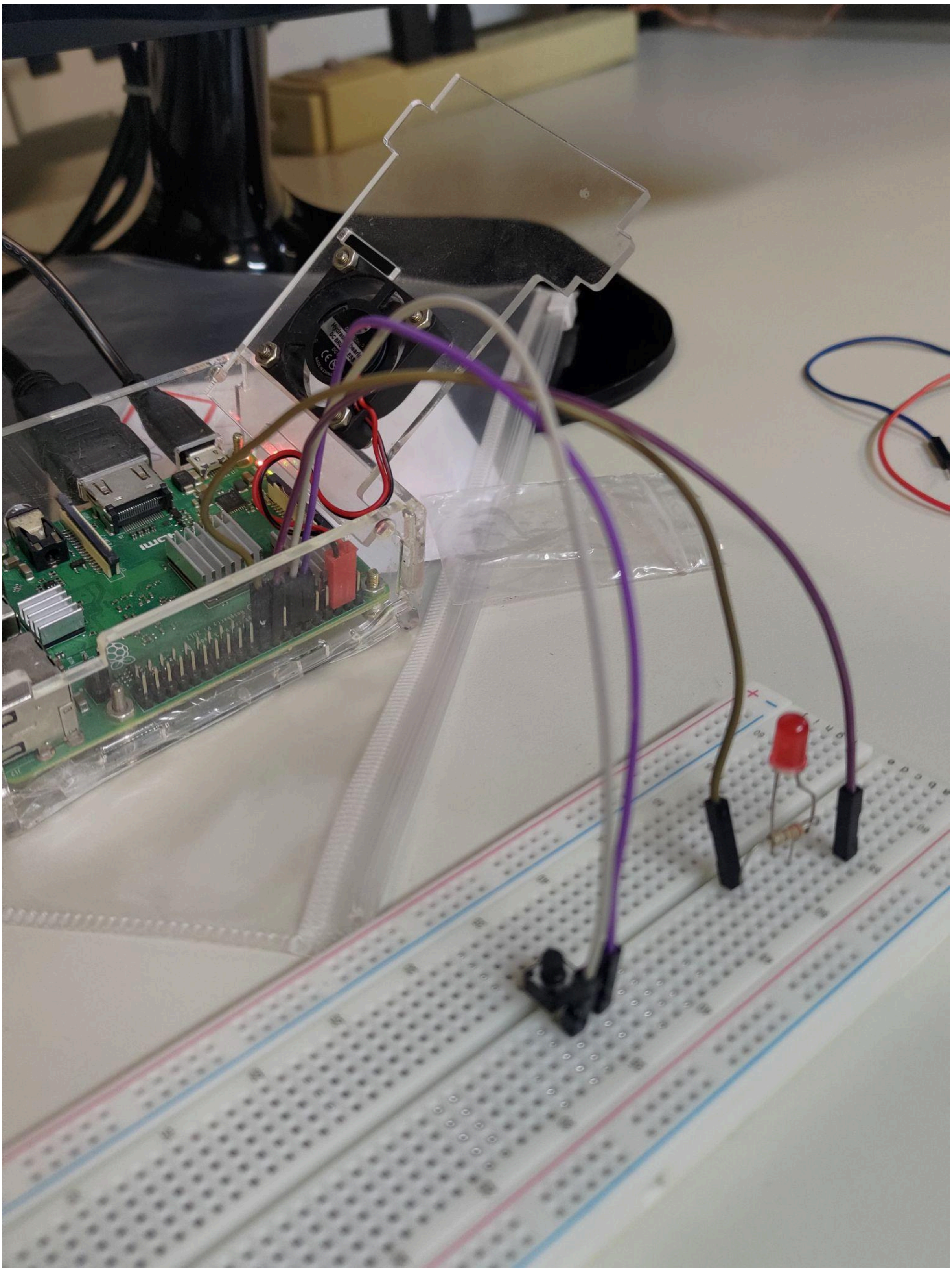

Trigger
Measure
Waveform
Tools
File
Vertical

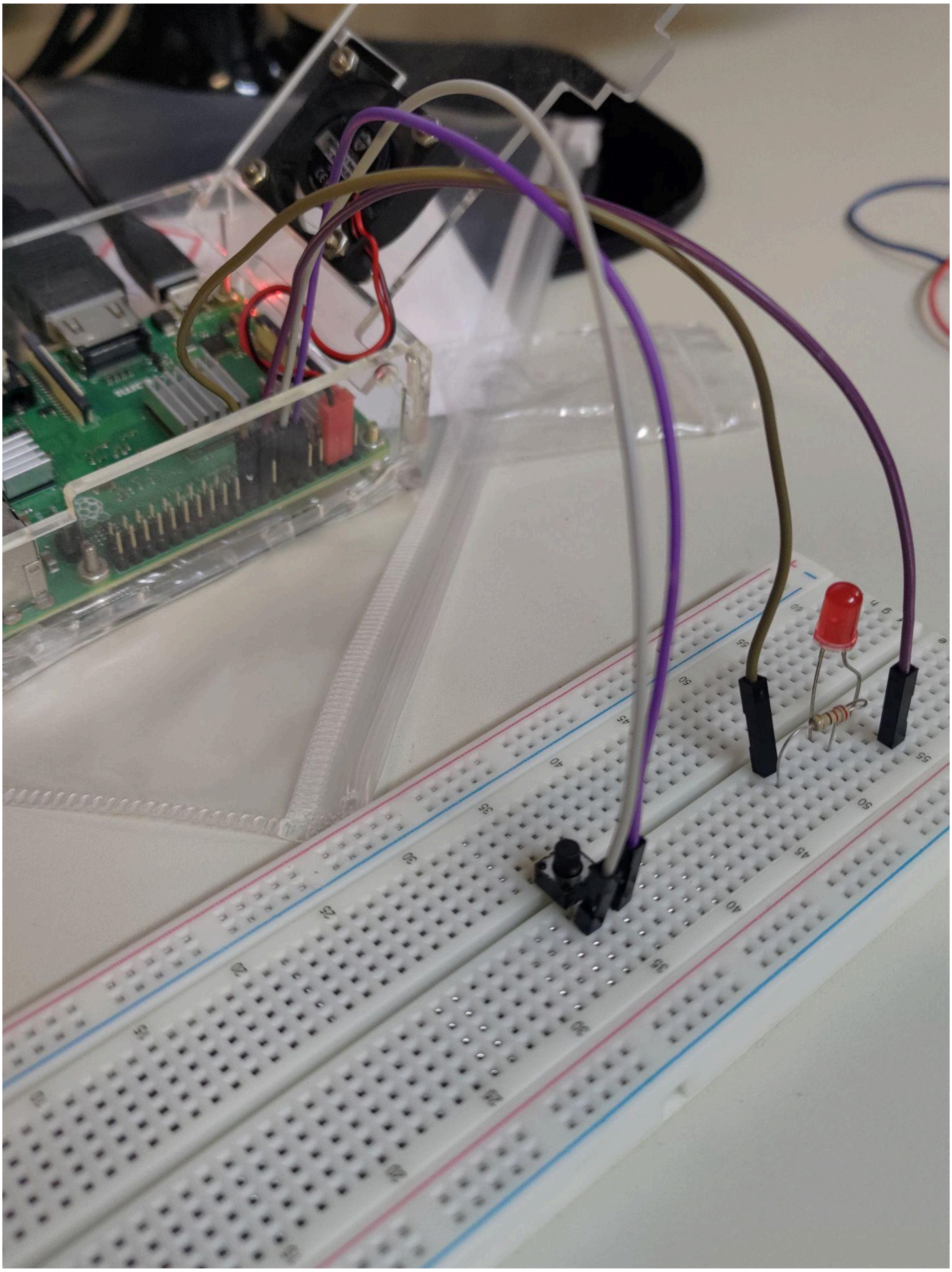

Agilent Technologies
InfiniiVision
DSO-X 2002A
70 MHz
KEYSIGHT TECHNOLOGIES
Aquisição
Normal
1.00MSa/s
Canais
Medições
1.52V
100.07Hz
3.17V
-160mV
LimBW
Fina
Invertida
Ponta de prova
python3 pwm.py

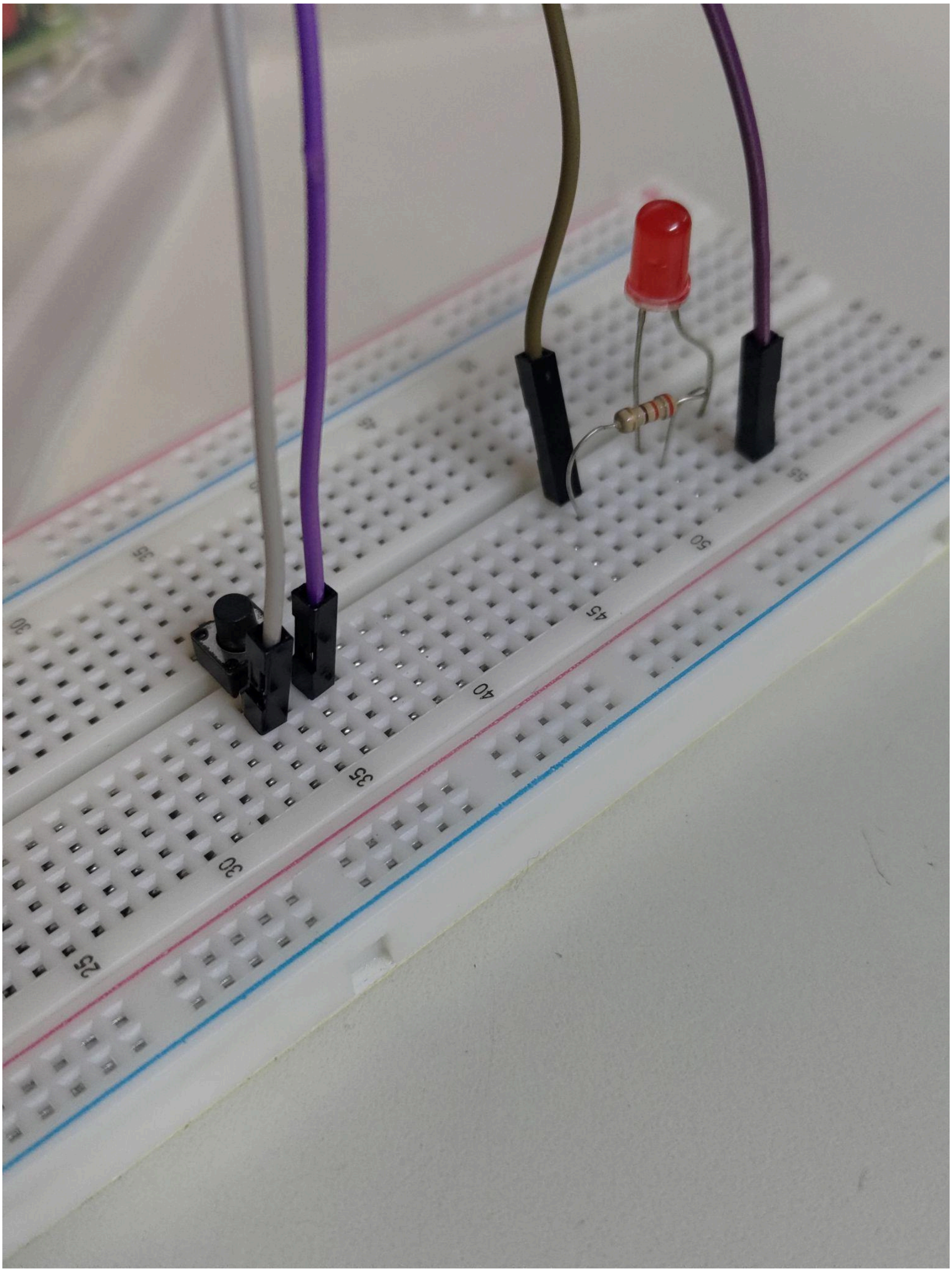


IMAGENS DA MONTAGEM DE botao.py









funcionamento de LED PWM antes de modificar o código para o formato em [PWM.py](PWM.py):
<https://youtube.com/shorts/qyYfvTzAD2E?feature=share>